

SPOKANE 17 SSW, WA, USA

WMO: 742040

Lat: 47.4174N

Lon: 117.5264W

Elev: 2267

StdP: 13.53

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 04136

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.1	7.3	-5.3	4.5	1.9	0.9	6.3	10.1	12.6	41.1	11.2	39.3	0.5	N/A	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.6	92.8	61.9	89.7	61.3	86.3	60.5	64.9	85.6	63.3	84.9	61.8	83.2	4.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	78.3	71.3	55.4	71.1	67.8	53.3	65.8	66.7	31.0	85.7	29.7	84.8	28.6	82.9	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.5	8.4	DB	-11.0	98.2	7.9	3.5	-16.7	100.8	-21.3	102.9	-25.7	104.9	-31.5	107.4	(4)
(5)			WB	-8.6	67.7	7.3	2.3	-13.9	69.3	-18.1	70.6	-22.2	71.9	-27.5	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	28.3	30.7	37.8	44.1	52.5	57.8	64.8	64.1	55.1	44.0	34.7	27.8
	DBStd	14.86	9.45	8.76	6.96	5.94	6.49	6.14	5.47	5.41	6.44	7.10	8.56	8.64
	HDD50	3223	672	541	378	195	49	6	0	0	23	210	460	689
	HDD65	7356	1137	961	842	627	390	227	74	83	302	650	909	1154
	CDD50	1482	0	0	1	18	126	241	458	436	176	25	1	0
	CDD65	138	0	0	0	0	1	12	66	54	5	0	0	0
	CDH74	5951	0	0	0	15	155	582	2360	2225	596	18	0	0
	CDH80	2482	0	0	0	0	21	200	1050	1008	200	3	0	0
Wind	WSAvg	3.4	3.1	3.6	4.1	4.4	3.8	3.6	3.1	2.9	2.9	2.9	3.0	3.0
Precipitation	PrecAvg	16.1	2.2	1.4	1.4	1.3	1.5	1.2	0.4	0.4	0.5	1.4	2.1	2.4
	PrecMax	23.8	4.6	3.3	2.7	2.7	2.6	2.3	2.8	1.4	1.4	3.4	4.3	5.0
	PrecMin	11.7	1.3	0.2	0.3	0.5	0.3	0.2	0.0	0.0	0.0	0.3	0.5	1.1
	PrecStd	2.6	0.8	0.7	0.7	0.6	0.6	0.5	0.6	0.4	0.4	0.8	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	53.5	63.2	76.5	82.9	92.5	96.1	97.6	90.4	75.8	60.2	52.0
		MCWB	43.1	45.3	47.9	55.2	59.6	63.9	62.7	61.9	60.6	55.8	50.3	46.5
	2%	DB	45.3	49.1	57.7	69.1	78.3	86.0	92.6	93.3	85.8	68.9	55.1	46.4
		MCWB	40.4	42.9	46.0	52.0	58.0	62.3	62.0	61.7	59.1	52.5	48.4	42.4
	5%	DB	42.3	46.0	53.5	63.1	74.6	80.1	89.7	89.9	81.2	63.9	51.3	42.5
		MCWB	38.7	41.2	44.4	49.1	56.6	59.3	61.4	60.9	57.5	50.1	45.3	39.2
	10%	DB	39.6	42.8	49.9	58.2	70.1	76.2	86.5	85.9	76.3	59.0	48.0	39.7
		MCWB	36.8	38.8	42.9	46.3	54.0	57.5	60.9	59.8	55.8	48.0	43.6	37.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	46.4	51.0	58.5	61.6	66.7	68.4	65.2	64.2	58.1	52.4	46.7
		MCDB	48.6	52.3	58.1	74.1	77.7	89.5	87.8	89.9	82.3	69.8	58.5	50.8
	2%	WB	40.7	44.8	48.2	53.3	59.5	64.0	65.9	63.4	61.3	53.9	49.5	43.0
		MCDB	43.9	49.4	55.4	66.2	75.3	83.9	84.5	86.8	80.0	63.9	54.9	46.3
	5%	WB	38.6	42.4	46.3	50.6	57.9	61.5	63.7	62.2	58.9	51.9	46.5	39.9
		MCDB	41.2	45.8	53.2	62.9	72.8	79.4	85.3	85.9	79.3	61.0	50.6	42.5
	10%	WB	36.5	39.7	44.3	48.1	55.8	59.4	62.0	60.9	56.6	49.7	44.0	37.4
		MCDB	38.9	43.0	50.5	57.8	69.0	74.7	83.6	84.1	74.3	57.6	47.8	39.9
Mean Daily Temperature Range	5% DB	MDBR	13.5	16.2	19.9	23.0	27.4	28.6	37.6	38.6	36.0	27.6	18.1	13.3
		MCDBR	14.4	19.5	26.3	33.2	35.8	38.9	44.7	47.2	46.7	36.9	22.8	13.8
		MCWBR	11.5	13.1	16.4	17.6	17.3	17.3	18.0	19.3	22.2	21.8	15.0	11.0
	5% WB	MCDBR	14.2	16.0	22.9	31.1	32.3	36.7	38.7	41.6	40.2	29.6	19.4	12.7
		MCWBR	11.3	11.2	15.2	16.8	15.8	16.5	16.2	17.6	19.6	18.2	13.9	10.5
Clear-Sky Solar Irradiance	taub	0.270	0.278	0.291	0.316	0.329	0.330	0.330	0.339	0.316	0.301	0.279	0.268	
	taud	2.505	2.550	2.563	2.471	2.455	2.478	2.473	2.433	2.488	2.533	2.555	2.503	
	Ebn at Noon	266	288	299	298	297	296	294	287	285	274	260	252	
	Edh at Noon	19	23	27	32	34	34	33	33	29	23	18	17	
All-Sky Solar Radiation	RadAvg	340	633	973	1451	1861	2017	2354	1952	1395	832	401	266	
	RadStd	41	85	111	138	148	171	122	127	128	91	42	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+85

Nomenclature: See separate page